Elmo's Cookie Waves

Euler's Equation Makes Waves, and What Phase Means

November 4, 2026 · Nathaniel

Name: _____ Date: _____

The story

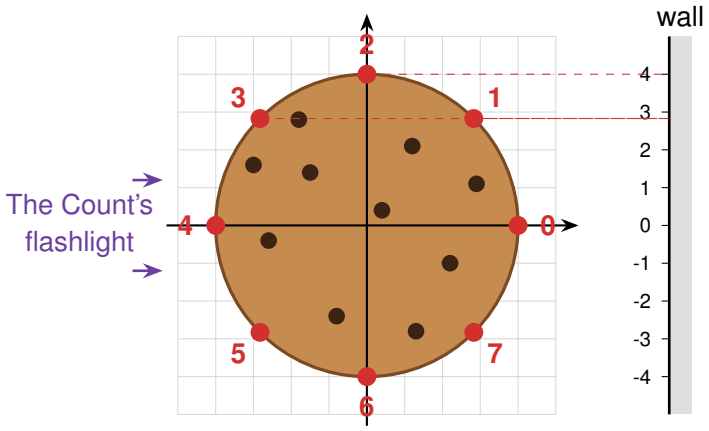
In The Cookie Circle, **Elmo** walked around the edge of Cookie Monster's giant cookie. Now Elmo keeps walking, around and around, at the same speed. **The Count** shines a flashlight from the side, so Elmo's **shadow** moves up and down on the wall.

"Elmo is going around and around. But Elmo's shadow goes up and down!"

That up-and-down shadow is a **wave**. Radios, phones, Wi-Fi, light, and sound are all made of waves like this. Let's draw one.

Elmo's shadow on the wall

The cookie is drawn on a grid. Its edge is **4 squares** from the middle. Elmo starts at spot **0** and walks around to the left (up first), stopping at each numbered spot. Each spot is **one eighth** of a lap. The dashed lines show his shadow's height on the wall.



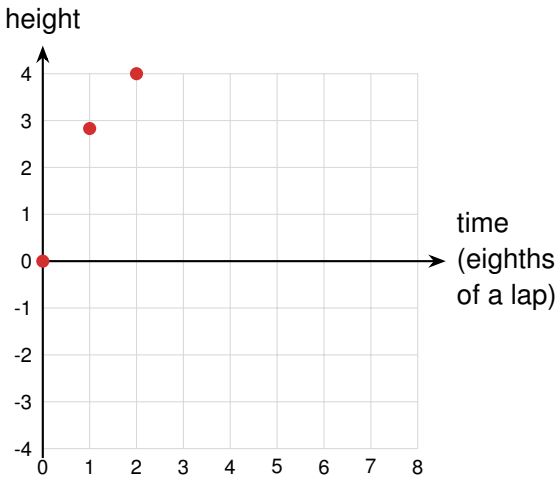
Question 1. Fill in Elmo's shadow height at each spot. Use the grid and the wall. (Spots 1, 3, 5 and 7 are a little less than 3 squares, so write "about 3.")

Spot	0	1	2	3	4	5	6	7	8
Height	0	3	4						

Spot 8 is back at the start: one whole lap!

Draw the wave

Question 2. Put a **red** dot for each height from your table: spot number across, height up. Then connect the dots with a smooth, curvy line.



Going **around** a circle makes a **wave** that goes **up and down**! This wave has a name: the **sine wave** (say "sign"). Every lap around the cookie draws one more bump up and one more dip down.

Three words engineers use

Word	What it means for Elmo
Amplitude	How **tall** the wave is. It is the size of the cookie: how far the edge is from the middle.
Frequency	How many **laps** Elmo walks every second. The Count counts them: *"ONE lap! TWO laps! THREE laps in one second! Ah ah ah!"* That is 3 **hertz**.
Phase	**Where on the circle** you are, or how far **ahead** one walker is of another. We measure it in turns, or in degrees: a whole turn is $360°$.

Question 3. The wave's highest point is _____ squares. So its amplitude is _____.

Question 4. A radio station at 100 megahertz makes its wave go around **100 million** laps every second! If Elmo walks 5 laps in one second, his frequency is _____ hertz.

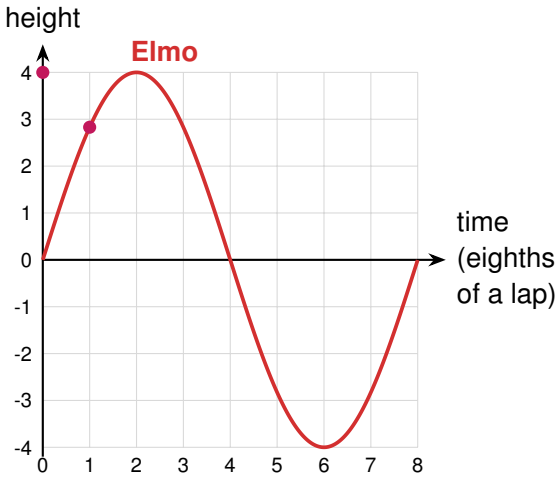
Phase: Abby starts ahead

Abby Cadabby wants to walk too. But Abby starts at spot **2**, at the **top** of the cookie, a **quarter turn ahead** of Elmo. They walk at the same speed.

Question 5. Fill in Abby's height when Elmo is at each spot. (Abby is always 2 spots ahead.)

Elmo's spot	0	1	2	3	4	5	6	7	8
Abby's height	4	3							

Question 6. Draw Abby's wave in **pink** on the grid. Elmo's wave is already there.



Question 7. Abby's wave looks just like Elmo's, but **slid over**. Whose wave reaches the top first?

_____

Abby's wave is the same wave, **shifted**. That shift is called the **phase**. Abby is a quarter turn ahead, so her phase is $90°$ ahead of Elmo's. Engineers say the two waves are $90°$ **out of phase**.

Question 8. Fill in the phase for each turn.

How far ahead	Phase
quarter turn	$90°$
half turn	
three-quarters turn	
whole turn	

Oscar's opposite wave

Oscar the Grouch starts at spot **4**, a **half turn** $(180°)$ ahead. When Elmo's shadow goes up, Oscar's goes down. *"Scram!"*

Question 9. Fill in Oscar's heights, then **add** Elmo's and Oscar's heights together.

Spot	0	1	2	3	4	5	6	7	8
Elmo	0	3	4	3	0	-3	-4	-3	0
Oscar	0	-3							
Elmo + Oscar	0	0							

Question 10. What happens when you add two waves that are a half turn apart?

_____

Waves a half turn apart **cancel**: together they make **nothing**! Waves that start together (**in phase**) **add up** to a bigger wave. This is called **interference**.

Question 11. Elmo's twin walks right next to him, in phase. Add their heights at spot 2: $4 + 4 =$ _____. The new wave is _____ (taller or shorter?).

How engineers use phase

- **Noise-canceling headphones** listen to the noise and play the same wave a half turn ahead, so the noise cancels.
- **Antennas** working together add their waves in phase in one direction, so the radio hears farther, and cancel in other directions.
- **Radios** send messages by changing the phase of a wave: a little turn ahead or behind means a different message.

Back to Euler's equation

The Count's Math Club: Euler's wave equation

In The Cookie Circle, $e^{j \times \text{steps}}$ meant "walk that many steps around the circle." Engineers write the steps as θ (the Greek letter "theta"). Euler showed:

$$e^{j\theta} = \underbrace{\cos\theta}_{\text{floor shadow}} + j \underbrace{\sin\theta}_{\text{wall shadow}}$$

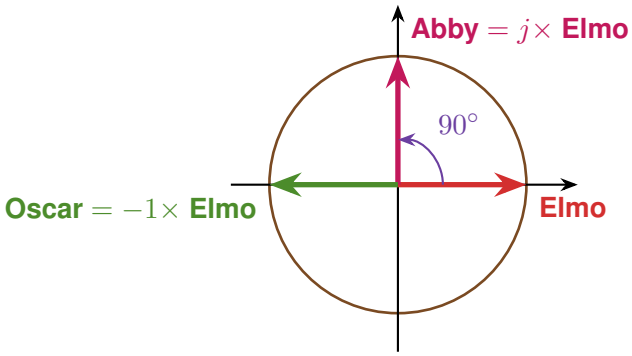
The **wall shadow** is the sine wave you drew. The **floor shadow** (a flashlight from above) is the **cosine wave**, which is Abby's wave! Radio engineers call these two shadows **I** and **Q**.

The big secret: a phase shift is a turn you multiply by. Abby is a quarter turn ahead of Elmo. A quarter turn is j! So

$$\text{Abby} = j \times \text{Elmo}.$$

Oscar is a half turn ahead: Oscar $= -1 \times$ Elmo. That is why his wave is upside down, and why Elmo + Oscar $= 0$.

Question 12. Which number do you multiply by to shift a wave a quarter turn? _____ A half turn? _____



Engineers draw each wave as an **arrow** on the cookie, called a **phasor**. The arrow's **length** is the amplitude. The arrow's **direction** is the phase. As time goes by, all the arrows spin around together, like Elmo, Abby, and Oscar walking. One arrow tells you everything about a wave!

Draw it!

Draw a cookie with three arrows for Elmo, Abby, and Oscar. Next to it, draw their three waves.

For teachers: Elmo's position after angle $\theta = \omega t$ is $4e^{j\omega t}$; his wall shadow is $4\sin\omega t$ (the quadrature, Q, component) and the floor shadow is $4\cos\omega t$ (in-phase, I). The heights at the eighth-turn spots are $4\sin 45° \approx 2.83$, written "about 3." Abby's wave $4\sin(\omega t + 90°) = 4\cos\omega t$ equals the imaginary part of $j \cdot 4e^{j\omega t}$: a phase shift ϕ is multiplication of the phasor by $e^{j\phi}$. Oscar's $\phi = 180°$ gives $e^{j\pi} = -1$ and destructive interference. The radio examples are noise cancellation, phased arrays, and phase-shift keying. Sesame Street characters appear by name only, for classroom use; this worksheet is not affiliated with or endorsed by Sesame Workshop.

Answer key

Question	Answer
1	0, 3, 4, 3, 0, -3, -4, -3, 0 (the 3s are about 2.83)
2	One smooth bump up (spots 0 to 4) and one dip down (spots 4 to 8)
3	4; 4
4	5 hertz
5	4, 3, 0, -3, -4, -3, 0, 3, 4
6	Same shape as Elmo's, starting at the top
7	Abby's
8	Half turn: $180°$; three-quarters turn: $270°$; whole turn: $360°$
9	Oscar: 0, -3, -4, -3, 0, 3, 4, 3, 0. Elmo $+$ Oscar: 0 at every spot
10	They cancel: the sum is always 0
11	8; taller (twice as tall)
12	j; -1